

DIVYESH K

Chennai, Tamil Nadu, India | divyeshkamalanaban@gmail.com | +91 8667072079
linkedin.com/in/divyesh-kamalanaban | github.com/Divyesh-Kamalanaban

SUMMARY

Software Engineer with hands-on experience building full-stack web applications, machine learning pipelines, and optimized backend systems. Currently pursuing a B.Tech in Electronics and Communication Engineering with a **9.21 CGPA**, combining strong academic performance with practical problem-solving skills. Adept at Agile development practices, performance optimization, and integrating **AI/ML concepts** into functional prototypes to solve meaningful business challenges.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering 2023 – 2027
SRM Institute of Science and Technology (Ramapuram), Tamil Nadu, India
• Current CGPA: **9.21 / 10**

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Frontend & Web: React.js, Next.js, Redux Toolkit, Context API, Tailwind CSS, Material UI (MUI)

Backend & Cloud: Node.js, Express.js, AWS, Cloudflare Workers, REST APIs, MongoDB, SQLite, Prisma ORM

AI / ML Skills: TensorFlow, TensorFlow Lite Micro, scikit-learn, Random Forests, Transformers, Pandas, NumPy

Testing & Visualization: k6, Grafana, API Testing, Selenium, Unit Testing (pytest, Jest)

Productivity & Tools: Microsoft Word, Excel, PowerPoint, Git, Agile/Scrum, Linux

EXPERIENCE

Software Developer Intern | Circle Communications May 2025
Chennai, TN

- Built a React frontend with MUI components, Redux Toolkit for state management, React Context API for real-time data synchronization, and code-split routing using lazy loading for performance optimization.
- Generated structured PDF invoices averaging ~2ms per document via PDFKit with streamed HTTP delivery.
- Identified unbounded query issue returning full dataset on every request; implemented server-side pagination reducing p95 response time from 1.97s to 19ms and sustaining 900+ req/sec across 1,000 concurrent users with 0% error rate.
- Analyzed requirements and contributed to technical documentation for ongoing projects, while participating in peer code reviews to uphold coding standards.

Summer Research Intern (ML & Power Systems) | National Institute of Technology (NIT) Trichy
Jun 2025 – Jul 2025

Trichy, TN

- Designed and implemented a modular fault detection engine for CIGRE-MV power networks, managing requirement scoping, iterative prototyping, and validation.
- Architected a multi-tier **random forest ensemble** in Python using Object-Oriented principles, achieving 92% fault detection and 100% fault localization accuracy.
- Synthesized 21,000+ labeled power grid scenarios using pandapower to build a robust supervised learning dataset, improving model accuracy to 95% across diverse load conditions.
- Replaced Newton–Raphson solver with an optimized DLPF-based implementation, reducing voltage computation time by 70% to improve real-time inference capabilities.

PROJECTS

Sororine — Full-Stack Safety Intelligence Platform

Jan 2026 – Feb 2026

- Architected and deployed a full-stack web platform using Next.js App Router and TypeScript, featuring API routes, an ORM-backed data layer (Prisma/SQLite), and a stateless JWT-secured session model.
- Implemented a risk-scoring algorithm aggregating real-time incident data within a 2km geospatial radius to generate actionable safety metrics.

SleeQC — Adaptive Embedded Security Prototype on ESP32

Aug 2025 – Oct 2025

- Prototyped a resource-adaptive Post-Quantum Cryptography system on an edge-constrained ESP32-S3.
- Deployed a **TinyML decision engine** using **TensorFlow Lite** to monitor real-time system metrics and autonomously switch between security profiles.
- Managed complex C/C++ cross-language linker dependencies within the ESP-IDF build system, resolving component conflicts to achieve a stable, reproducible build pipeline.

CERTIFICATIONS

- GitHub Foundations